# TL Matter SDK Release Note

Telink Semiconductor

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# Contents

# Acknowledgements

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## Revision History

- **V1.1.0**: Initial release by lingzhi.
- **V1.1.1**: Support the dual-mode switch of zigbee and matter by fengtai.

# 1 Overview

This document will introduce a matter customer solution based on our TLSR952x(2MB or 4MB), TL321x(2MB) and TL721x(2MB) platform.

Matter v1.4 was officially released in March 2025, and the certification process for version 1.4 has also commenced. Telink has been closely monitoring the updates to Matter, actively participating in all TE, SVE, and Plugfest events organized by the CSA.

In this document, we will provide detailed recommendations for Telink customers regarding the preferred code versions for development. Additionally, we will furnish specific results from internal testing, including certification tests and ecosystem tests.

> Note: All the tests based on Telink TLSR952x(2MB or 4MB), TL321x(2MB) and TL721x(2MB) platform.

# 2 Code Bases

## 2.1 Code Repository

Our TL_Matter SDK is a Modified Matter SDK, and that it supports the dual-mode of zigbee and matter.

A complete Matter SDK consists of several repositories (Matter, Zephyr and so on). In this release, we recommend customers to use the following code bases for development on the Telink TLSR952x(2MB or 4MB) and TL321x(2MB) and TL721x(2MB) platform:

- **Matter repository**: https://github.com/project-chip/connectedhomeip/tree/master

  **Commit ID**: af29048a0042a529e0ec37f575e09b41b08ce438

  **Patch**: tl_matter_sdk_matter_v1_1_1.patch

- **Zephyr repository**: https://github.com/telink-semi/zephyr/tree/develop_v3.3

  **Commit ID**: e20d103aa29b2ea9febdff81d16cf7336c3721d0

  **Patch**: tl_matter_sdk_zephyr_v1_1_1.patch

- **hal_telink repository**: https://github.com/telink-semi/hal_telink/tree/develop

  **Commit ID**: 541f19515a7796411ed68352ffd9718472212ca6

  **Patch**: tl_matter_sdk_hal_v1_1_1.patch

- **mcuboot repository**: https://github.com/mcu-tools/mcuboot/tree/main

  **Commit ID**: 1558e7ab0aadb4eac11f03befb5ccd3fa3f0aafe

  **Patch**: tl_matter_sdk_mcuboot_v1_1_1.patch

- **modules_lib_openthread repository**: https://github.com/zephyrproject-rtos/openthread/tree/main

  **Commit ID**: aabbee49c8c9721ed1ca22a7b326ae3a23aaae09

  **Patch**: tl_matter_sdk_lib_openthread_v1_1_1.patch

### 2.2 Apply Patches

Please follow the steps below strictly in the given order.

- Apply the Matter patch

  1. Copy the `tl_matter_sdk_matter_v1_1_1.patch` to the path of connectedhomeip folder.
  2. Apply this patch

```
cd ~/connectedhomeip
git apply tl_matter_sdk_matter_v1_1_1.patch
```

- Apply the zephyr patch

  1. Copy the `tl_matter_sdk_zephyr_v1_1_1.patch` to the path of zephyr folder.
  2. Apply this patch and update hal, mcuboot

```
cd ~/zephyrproject/zephyr
git apply tl_matter_sdk_zephyr_v1_1_1.patch
cd ..
west update
```

- Apply the hal patch and fetch lib file

  1. Copy the `tl_matter_sdk_hal_v1_1_1.patch` to the path of hal_telink folder.
  2. Apply this patch

```
cd ~/zephyrproject/modules/hal/telink
git apply tl_matter_sdk_hal_v1_1_1.patch
cd ../../..
west blobs fetch hal_telink
```

- Apply the mcuboot patch

  1. Copy the `tl_matter_sdk_mcuboot_v1_1_1.patch` to the path of mcuboot folder.
  2. Apply this patch

```
cd ~/zephyrproject/bootloader/mcuboot
git apply tl_matter_sdk_mcuboot_v1_1_1.patch
```

- Apply modules_lib_openthread patch

  1. Copy the `tl_matter_sdk_lib_openthread_v1_1_1.patch` to the path of lib_openthread folder.
  2. Apply this patch

```
cd ~/zephyrproject/modules/lib/openthread
git apply tl_matter_sdk_lib_openthread_v1_1_1.patch
```

## 2.3 Change Device Type

By default, the lighting-app is a dimmable light, if you need to change the device type to extended color light, just need to change it via ZAP tool.

1. Run the ZAP tool

```
cd ~/connectedhomeip
source scripts/activate.sh
./scripts/tools/zap/run_zaptool.sh examples/lighting-app/lighting-common/lighting-app.zap
```

2. Click Endpoint1 to see the device type

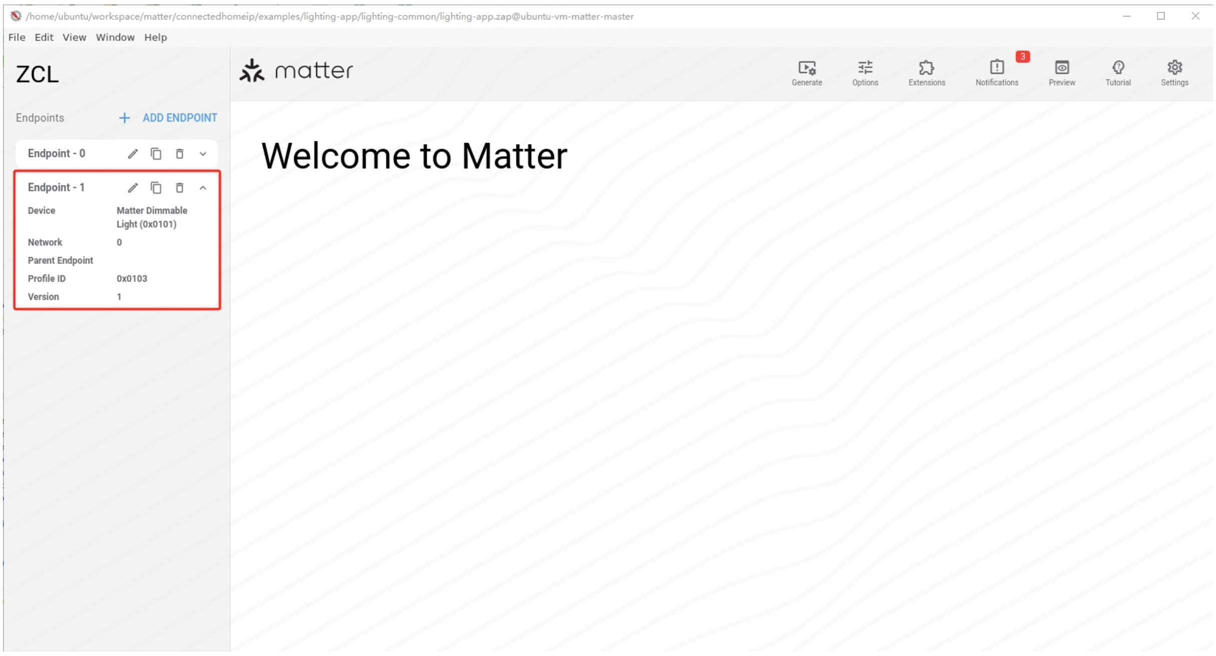



**Figure 1:** ZAP Endpoint1

3. Click `Edit` to change the device type in Endpoint1

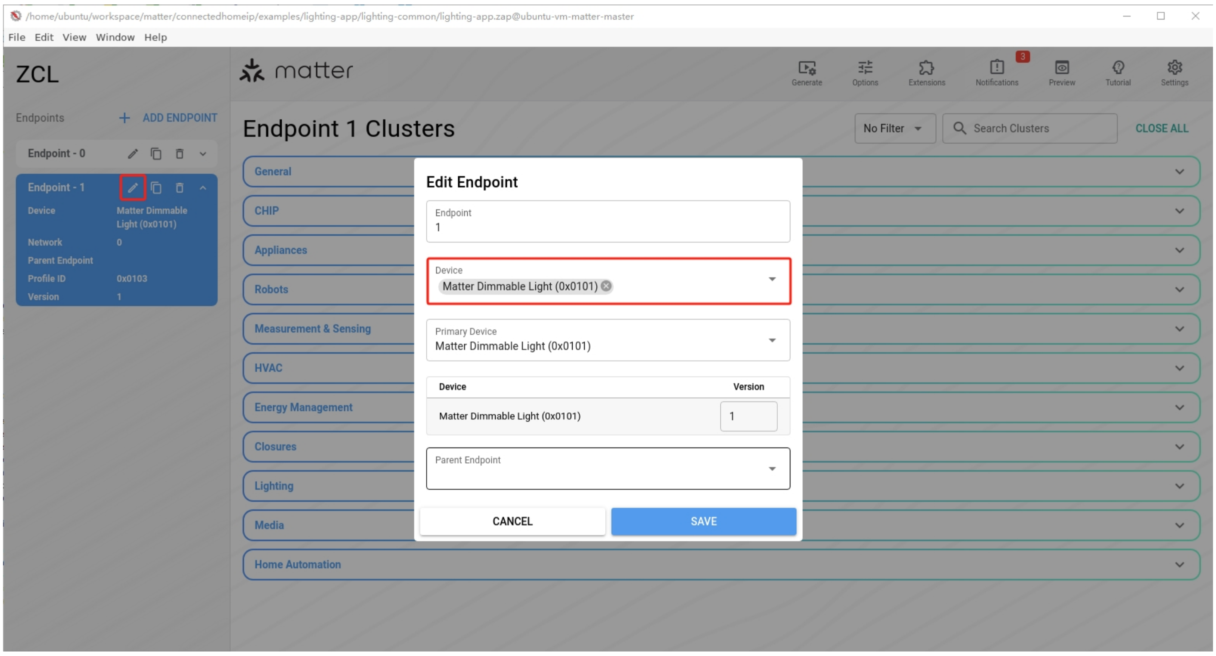



**Figure 2:** ZAP Change Device Type

4. Choose `Matter Extended Color Light(0x10D)`

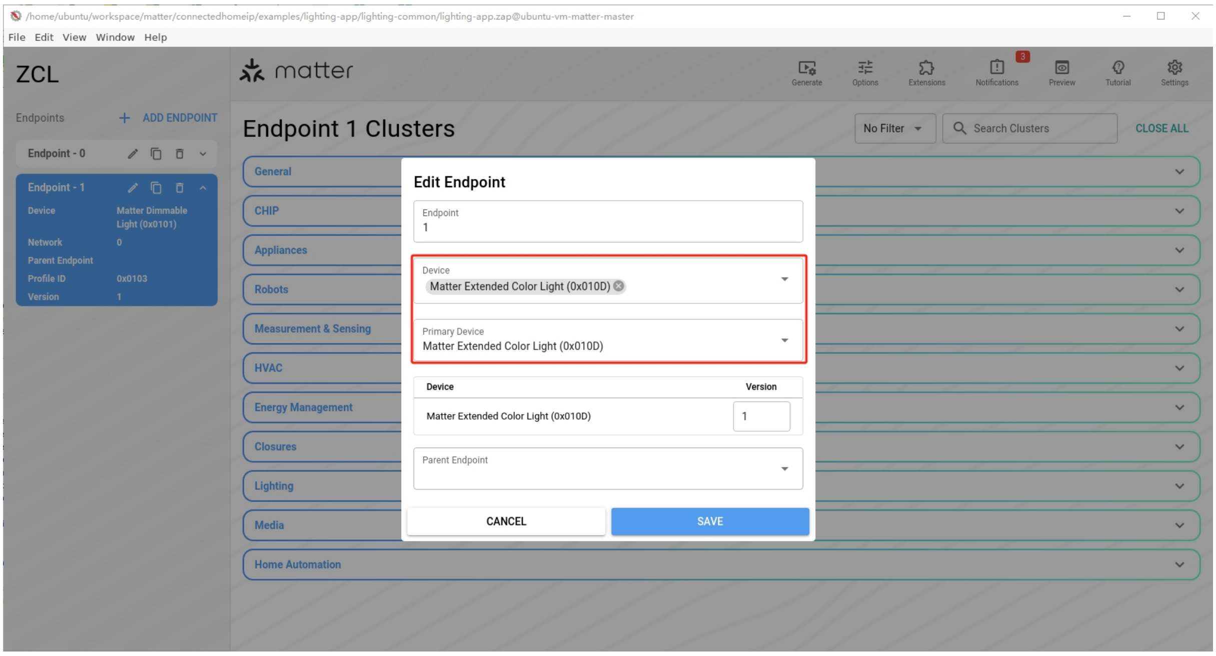



**Figure 3:** ZAP Choose Device Type

5. Save the changes

   Click `File-save` in the top left corner of ZAP page or use `Ctrl+S` directly to save the changes.

6. Update `lighting-app.matter` file

```
cd ~/connectedhomeip
./scripts/tools/zap/generate.py examples/lighting-app/lighting-common/lighting-app.zap
```

## 2.4 Generate the Firmware

1. Compile command:

```
cd ~/connectedhomeip
source scripts/activate.sh
cd examples/lighting-app/telink
rm -rf build/
west build -p auto -b tlsr9528a
west build -p auto -b tl3218x
west build -p auto -b tl7218x
```

```
cd examples/light-switch-app/telink
rm -rf build/
west build -p auto -b tlsr9528a_retention
west build -p auto -b tl3218x_retention
west build -p auto -b tl7218x_retention
```

   If you need to run in 4MB flash (default is 2MB flash),
   you can add `-DFLASH_SIZE=4m` to the compile command. such as:

```
west build -p auto -b tlsr9528a -- -DFLASH_SIZE=4m
```

   If you want to open dual-mode,
   you can add `-DCONFIG_DUAL_MODE=y` to the compile command. such as:

```
west build -p auto -b tlsr9528a -- -DCONFIG_DUAL_MODE=y
```

> For the dual-mode, you can check the detailed flash layout in Appendix I.
> Among the dual-mode versions currently released, TLSR952x only has a 4MB version, and TL321x and TL721x only have a 2MB version. If you need other flash layout for the dual-mode, please contact Telink Semiconductor Company.

If you want to get more information about how to set up a Matter environment, please check our Telink Wiki page and visit our developer's guide at the end of this webpage.

# 3 Matter Basic Functionality Test Results

We used the official TestHarness(version 2.11-beta1+fall2024) to conduct detailed tests on the basic functionality of the Matter device, including the following test items.

| Test | Telink lighting-app | Telink light-switch-app |
|---|---|---|
| Factory Reset | PASS | PASS |
| Commissioning | PASS | PASS |
| OnOff Control | PASS | PASS |
| Reconnection | PASS | PASS |
| OTA Update | PASS | PASS |

# 4 Matter Ecosystem Test Results

To simulate the everyday interactions users have with IoT devices, we conducted testing on specific device types within prominent commercial ecosystems such as Google, Apple, Samsung and Amazon.

Our simulated testing primarily includes the following tests:

1. **Commissioning Test** – Attempting commissioning operations using the same device types across the four ecosystems (Google, Apple, Samsung and Amazon).
2. **Interaction Test** – Controlling devices that have already been commissioned within each of the four ecosystems (Google, Apple, Samsung and Amazon).
3. **Stability Test** – Testing the duration for which devices that are already commissioned in each of the four ecosystems (Google, Apple, Samsung and Amazon) can maintain an online status.
4. **Reconnection Test** – Powering off devices that are already commissioned in each of the four ecosystems (Google, Apple, Samsung and Amazon), waiting for a certain period, and then powering them back on to test whether the devices can successfully reconnect.

## 4.1 Test Conditions

### 4.1.1 Telink Platform

We use Telink TLSR952x(2MB or 4MB), TL321x(2MB) and TL721x(2MB) EVB as the device under test, to test the behavior of different device types across four ecosystems.


**Figure 4:** TLSR952x-Development-Board-Front

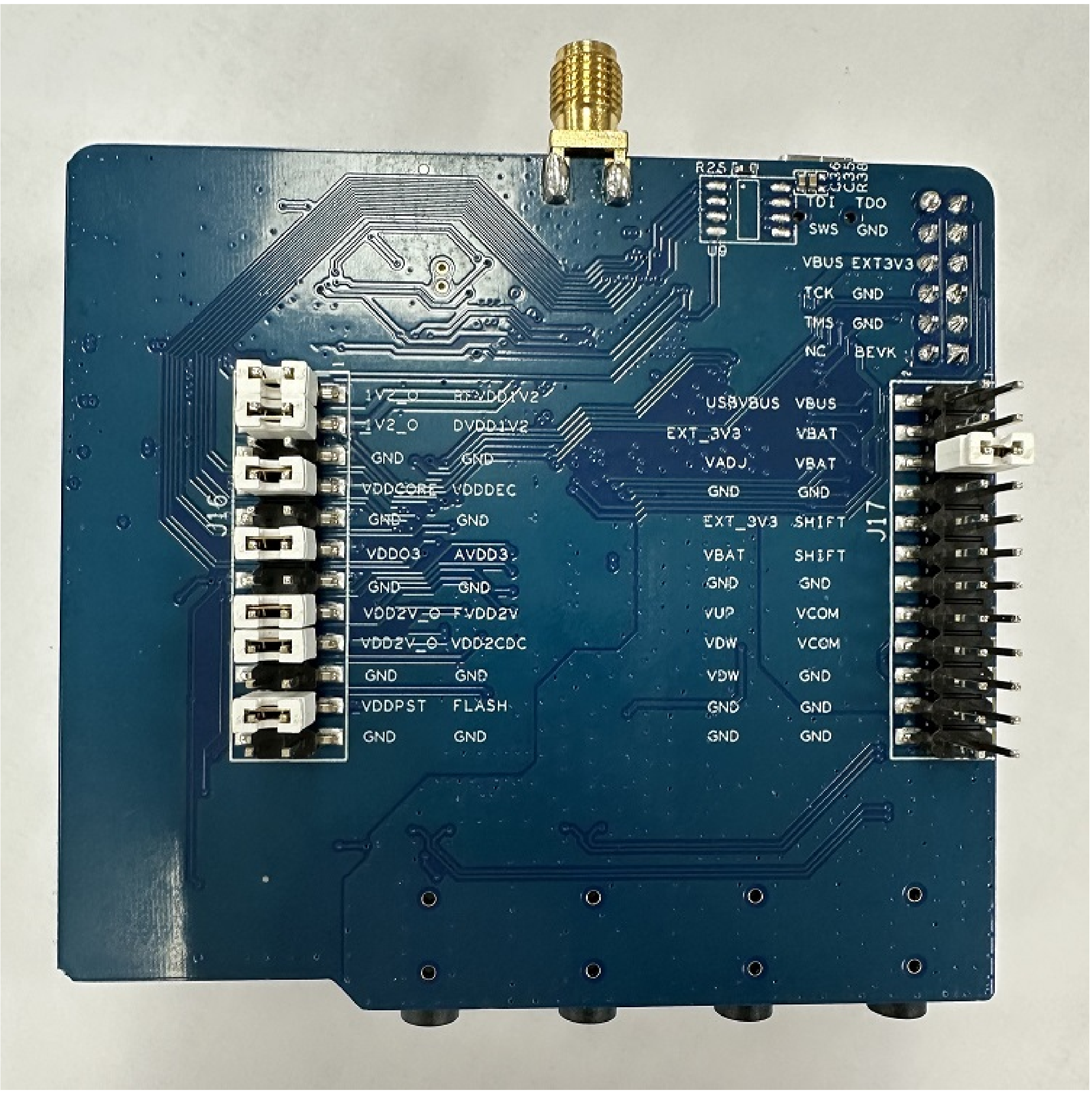



**Figure 5:** TLSR952x-Development-Board-Back

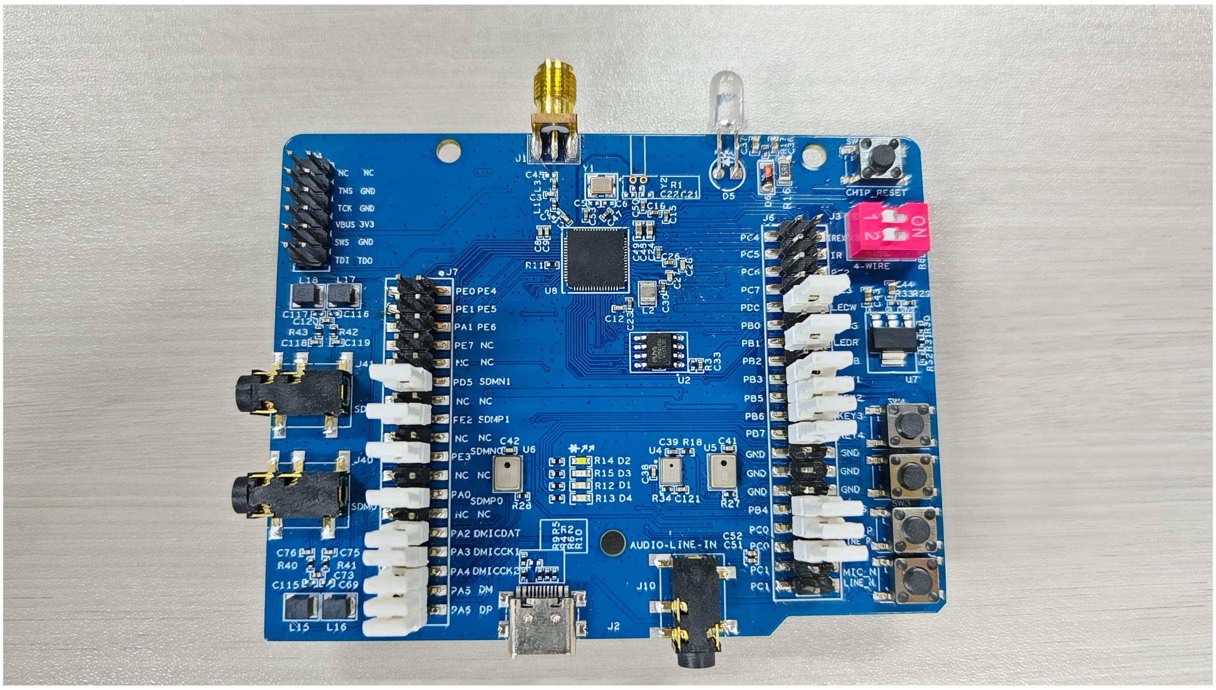



**Figure 6:** TL321x-Development-Board-Front

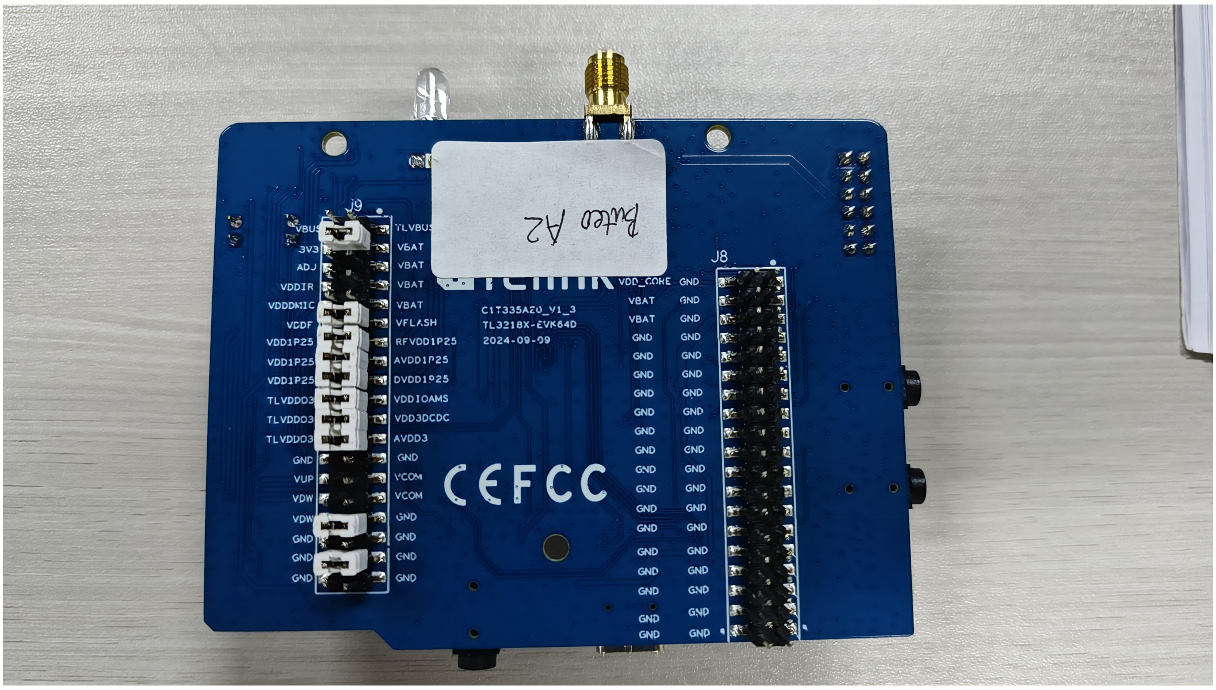



**Figure 7:** TL321x-Development-Board-Back

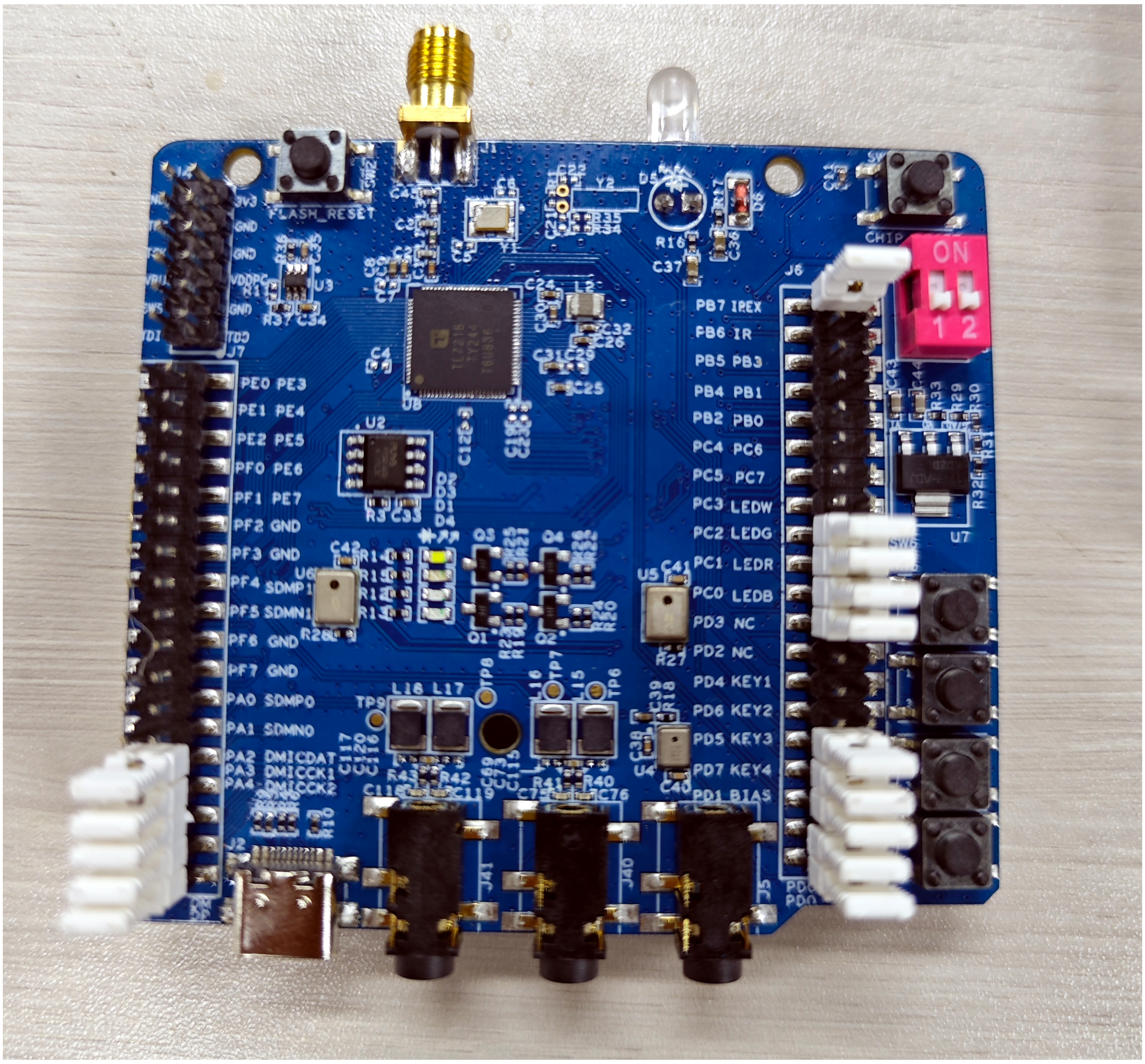
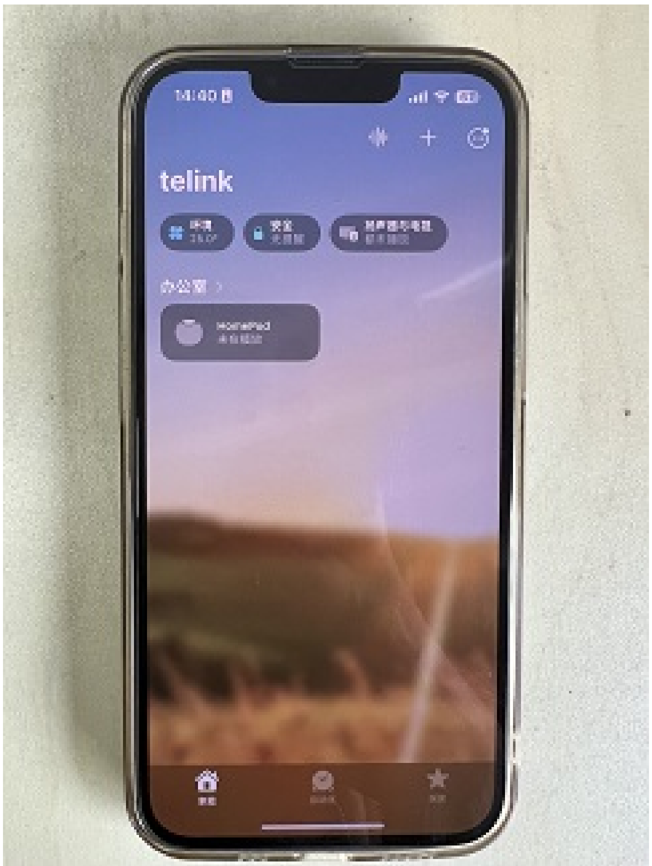



**Figure 8:** TL721x-Development-Board-Front


**Figure 9:** TL721x-Development-Board-Back

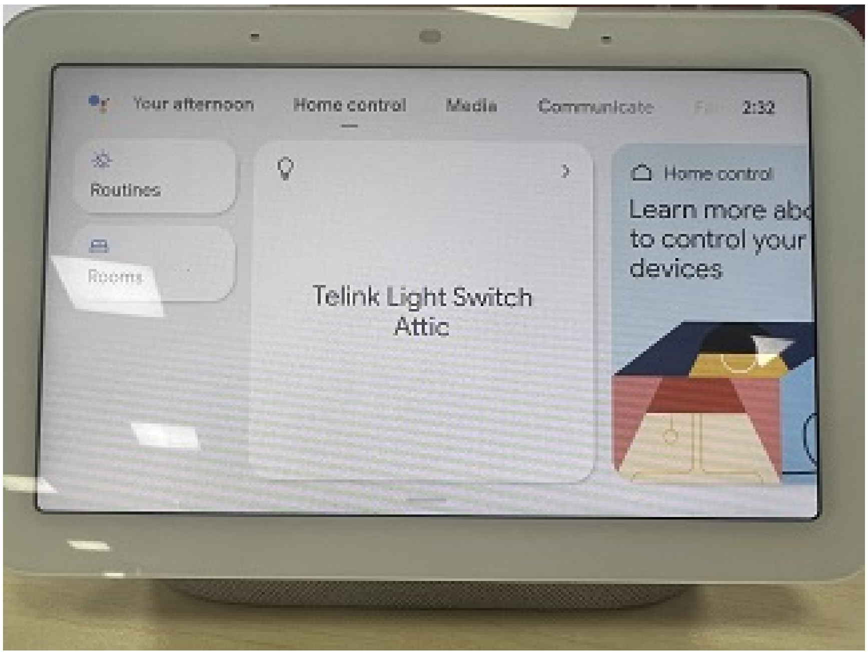
#### 4.1.2 Apple Ecosystem

- iPhone 13
  - iOS version: iOS 17.5.1 (21F90)
- Apple Homepod mini
  - software version: 17.5 (21L569)




#### 4.1.3 Google Ecosystem

- Pixel 6
  - Android version: 13
  - Google Play services: version 24.28.34 (190400-655627182)
- Google Nesthub (Gen 2)
  - Fuchsia Version: 18.20240225.3.63
  - Software Version: 58.102.2.613521407
  - Chromecast firmware version: 3.73.411796

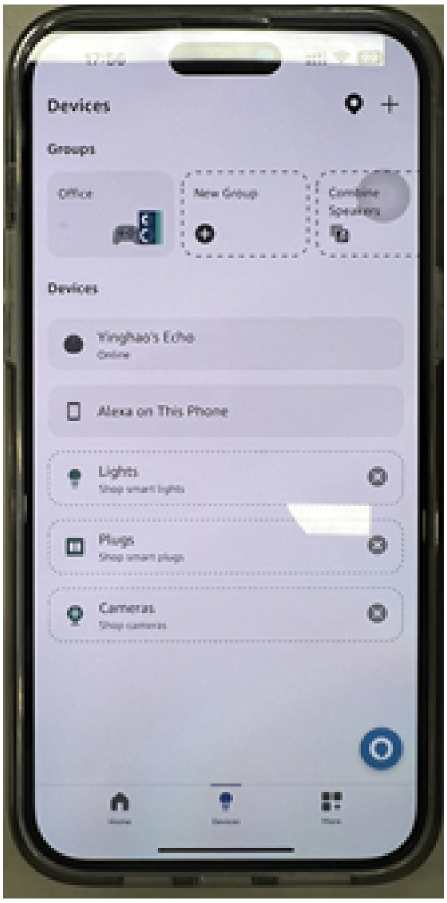



#### 4.1.4 Samsung Ecosystem

- Pixel 6
  - Android version: 13
- Samsung SmartThings Station
  - Firmware version: P9501ZCU1AXF5


### 4.1.5 Amazon Ecosystem

- iPhone 14
  - iOS version: iOS 17.5.1 (21F90)
- Amazon Alexa Echo 4
  - Device Software Version: 0010369673092




## 4.2 Matter Ecosystem Test

### 4.2.1 Matter Commissioning Test

Test procedure:

1. Commission devices into different ecosystems.
2. Remove the commissioned devices.
3. Do factory reset.
4. Repeat Step 1 and 2 a total of 10 times continuously.

Test results:

| Device Type | Google NestHub | Apple HomePod-mini | Samsung SmartThings Station | Amazon Alexa Echo |
|---|---|---|---|---|
| lighting-app | PASS | PASS | PASS | PASS |
| light-switch-app | PASS | PASS | PASS | PASS |

### 4.2.2 Matter Interaction Test

Test procedure:

1. Commission devices into different ecosystems.
2. Control the devices that has been commissioned into ecosystems. For example, with regards to a light, we would attempt to control its on/off using a mobile app within the respective ecosystems.

Test results:

| Device Type | Google NestHub | Apple HomePod-mini | Samsung SmartThings Station | Amazon Alexa Echo |
|---|---|---|---|---|
| lighting-app | PASS | PASS | PASS | PASS |

### 4.2.3 Matter Stability Test

Test procedure:

1. Commission devices into different ecosystems.

2. Leaving the commissioned devices in place for 7 days (7 * 24 hours) and periodically checking whether the devices can be reliably controlled during this period.

Test results:

| Device Type | Google NestHub | Apple HomePod-mini | Samsung SmartThings Station | Amazon Alexa Echo |
|---|---|---|---|---|
| lighting-app | On-Going | On-Going | On-Going | On-Going |
| light-switch-app | On-Going | On-Going | On-Going | On-Going |

#### 4.2.4 Matter Reconnection Test

Test procedure:

1. Commission devices into different ecosystems.
2. Power off the devices, then power them on again. Afterward, check whether these devices can reconnect to the networks or not.
3. Repeat Step 1 and 2 a total of 10 times continuously.

Test results:

| Device Type | Google NestHub | Apple HomePod-mini | Samsung SmartThings Station | Amazon Alexa Echo |
|---|---|---|---|---|
| lighting-app | PASS | PASS | PASS | PASS |
| light-switch-app | PASS | PASS | PASS | PASS |

#### 4.2.5 Multi Fabric Test

1. Apple Ecosystem Multi Fabric

   Test procedure:

   i. Commission the device into Apple HomePod-mini.

   ii. Share the device with Google, Samsung and Amazon ecosystems, make sure the device can be controlled by all four ecosystems.

| Device Type | Google NestHub | Samsung SmartThings Station | Amazon Alexa Echo |
|---|---|---|---|
| lighting-app | PASS | PASS | PASS |
| light-switch-app | PASS | PASS | PASS |

2. Google Ecosystem Multi Fabric

   Test procedure:

   i. Commission the device into Google Nesthub.

   ii. Share the device with Apple, Samsung and Amazon ecosystems, make sure the device can be controlled by all four ecosystems.

| Device Type | Apple HomePod-mini | Samsung SmartThings Station | Amazon Alexa Echo |
|---|---|---|---|
| lighting-app | PASS | PASS | PASS |
| light-switch-app | PASS | PASS | PASS |

3. Samsung Ecosystem Multi Fabric

   Test procedure:

   i. Commission the device into Samsung SmartThings Station.

   ii. Share the device with Apple, Google and Amazon ecosystems, make sure the device can be controlled by all four ecosystems.

| Device Type | Google NestHub | Apple HomePod-mini | Amazon Alexa Echo |
|---|---|---|---|
| lighting-app | PASS | PASS | PASS |
| light-switch-app | PASS | PASS | PASS |

4. Amazon Ecosystem Multi Fabric

   Test procedure:

   i. Commission the device into Amazon Alexa.

ii. Share the device with Apple, Google and Samsung ecosystems, make sure the device can be controlled by all four ecosystems.

| Device Type | Google NestHub | Apple HomePod-mini | Samsung SmartThings Station |
|---|---|---|---|
| lighting-app | PASS | PASS | PASS |
| light-switch-app | PASS | PASS | PASS |

# 5 Matter Certification Test Results

In order to bring Matter devices to market, it is imperative that they successfully undergo the Matter certification test, resulting in the issuance of a certificate by CSA. To achieve this, adherence to the latest Test Plans and test suites provided by the Matter community is mandatory.

To meet these requirements, we conducted a series of certification tests for the dual mode lighting app, selected a large number of automatic and python tests, and finally our device could pass all of these test cases. Here's the list of tested cases:

TestHarness version: 2.11-beta1+fall2024

| Test | Telink lighting-app |
|---|---|
| TC-ACE-1.1 | PASS |
| TC-ACE-1.2 | PASS |
| TC-ACE-1.3 | PASS |
| TC-ACE-1.4 | PASS |
| TC-ACE-1.5 | PASS |
| TC-ACE-1.6 | PASS |
| TC-ACE-2.1 | PASS |
| TC-ACE-2.2 | PASS |
| TC-ACL-2.1 | PASS |
| TC-ACL-2.2 | PASS |
| TC-ACL-2.3 | PASS |
| TC-ACL-2.6 | PASS |
| TC-CGEN-2.1 | PASS |
| TC-CGEN-2.4 | PASS |
| TC-CNET-1.3 | PASS |
| TC-CNET-4.6 | PASS |
| TC-DESC-2.2 | PASS |

| Test | Telink lighting-app |
|---|---|
| TC-DGGEN-2.3 | PASS |
| TC-DGGEN-2.4 | PASS |
| TC-DGGEN-3.1 | PASS |
| TC-DGTHREAD-2.2 | PASS |
| TC-DGTHREAD-2.3 | PASS |
| TC-DT-1.1 | PASS |
| TC-GRPKEY-2.1 | PASS |
| TC-IDM-1.2 | PASS |
| TC-IDM-1.4 | PASS |
| TC-IDM-10.1 | PASS |
| TC-IDM-10.3 | PASS |
| TC-IDM-11.1 | PASS |
| TC-LTIME-1.2 | PASS |
| TC-OPCREDS-3.1 | PASS |
| TC-SC-3.6 | PASS |
| TC-SC-5.1 | PASS |
| TC-SM-1.1 | PASS |
| TC-SM-1.2 | PASS |
| TC-ULABEL-2.1 | PASS |
| TC-ULABEL-2.2 | PASS |
| TC-ULABEL-2.3 | PASS |
| TC-ULABEL-2.4 | PASS |
| TC-G-2.1 | PASS |
| TC-I-2.1 | PASS |
| TC-OO-2.1 | PASS |
| TC-OO-2.3 | PASS |
| TC-OCC-2.1 | PASS |

| Test | Telink lighting-app |
| --- | --- |
| TC-OCC-2.3 | PASS |
| TC-RR-1.1 | PASS |
| TC-CC-4.1 | PASS |
| TC-CC-4.2 | PASS |
| TC-CC-4.3 | PASS |
| TC-CC-4.4 | PASS |
| TC-CC-5.1 | PASS |
| TC-CC-5.2 | PASS |
| TC-CC-5.3 | PASS |
| TC-CC-6.1 | PASS |
| TC-CC-6.2 | PASS |
| TC-CC-6.3 | PASS |
| TC-CC-7.1 | PASS |
| TC-CC-7.2 | PASS |
| TC-CC-7.3 | PASS |
| TC-CC-7.4 | PASS |
| TC-CC-9.1 | PASS |
| TC-CC-9.2 | PASS |
| TC-CC-9.3 | PASS |

# 6 Appendix I

The Flash Layout of dual-mode for TLSR952x(4MB), TL321x(2MB) and TL721x(2MB):

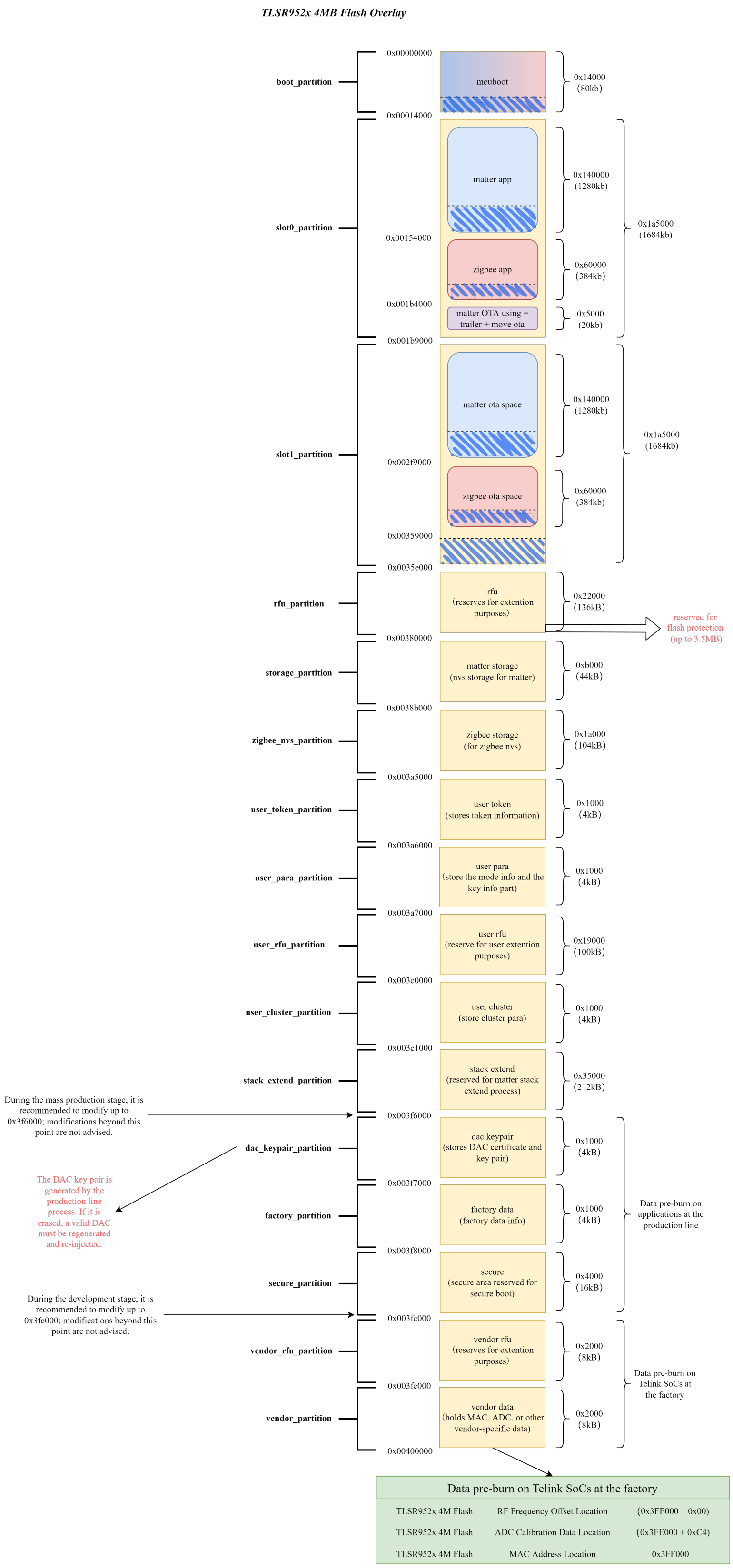
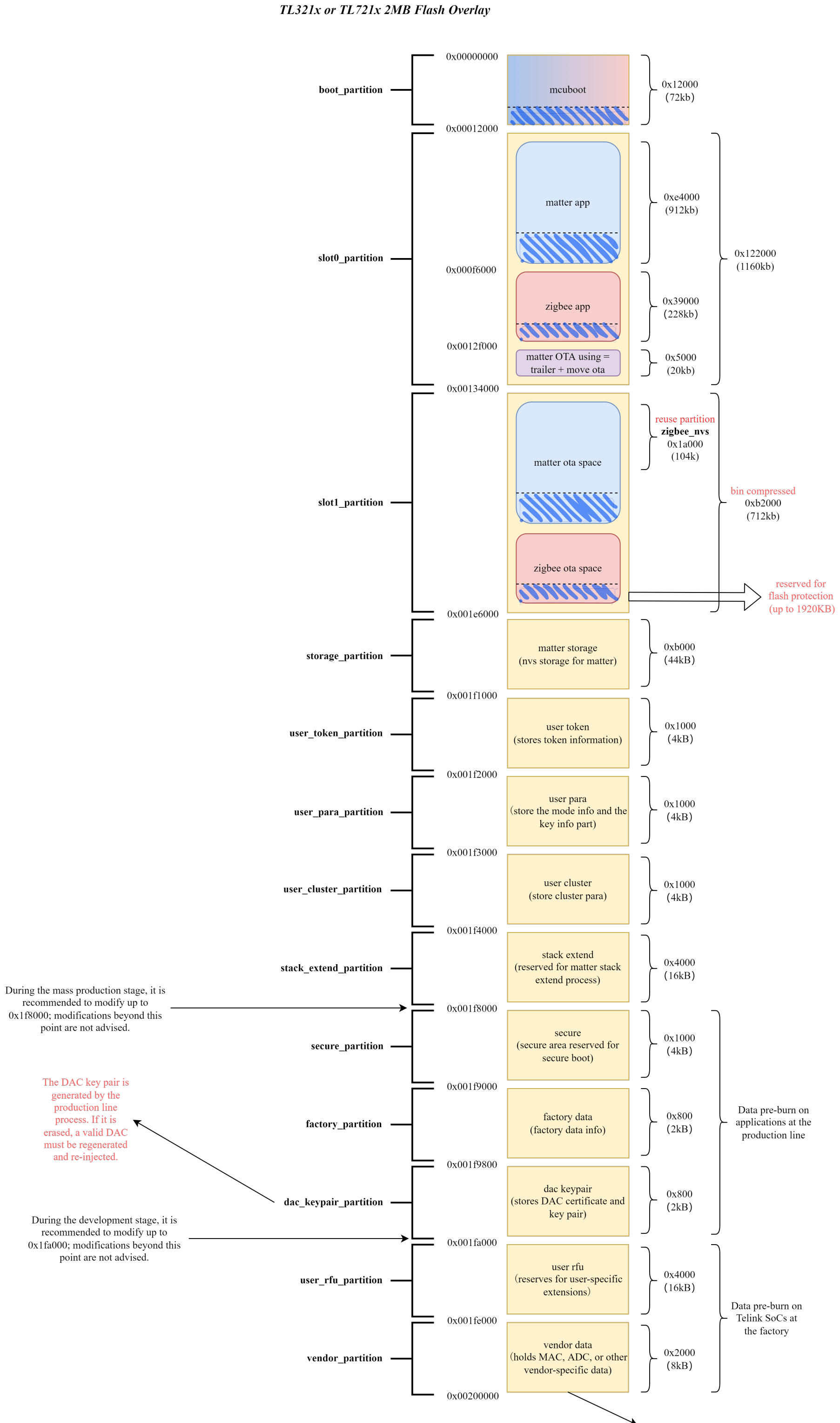



**Figure 10:** TLSR952x Dual-mode 4MB Flash Layout




| Data pre-burn on Telink SoCs at the factory | | |
|---|---|---|
| TL321x or TL721x 2M Flash | RF Frequency Offset Location | (0x1FE000 + 0x00) |
| TL321x or TL721x 2M Flash | ADC Calibration Data Location | (0x1FE000 + 0xC4) |
| TL321x or TL721x 2M Flash | MAC Address Location | 0x1FF000 |

**Figure 11:** TL321x and TL721x Dual-mode 2MB Flash Layout